Analysis for Lithium-Ion Battery Core Temperature Prediction using Physics Informed Machine Learning

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Problem Statement and Motivation

1. Real-time prediction of a battery's internal core temperature is critical for preventing thermal runaway, but directly measuring it remains physically impossible in commercial cells.
2. Traditional electrochemical models are too computationally heavy for real-time use, creating a major gap between complex physics and the lightweight processing requirements of a practical Battery Management System (BMS).
3. We aim to bridge this gap by using coupled electrical and thermal models to engineer precise ground-truth labels, enabling lightweight algorithms to accurately monitor battery health directly on embedded microcontrollers for vehicular safety.

Literature Review and Data Preparation

1. Our review of recent battery thermal management literature reveals a critical gap: commercial Battery Management Systems (BMS) rely on **simplistic surface temperature** [7], while highly accurate academic electrochemical models are too computationally expensive for real-time microcontroller deployment [8].
2. Current research heavily utilizes deep learning "black boxes" for state-of-health (SOH) estimation; however, these models often struggle to generalize across varying cell chemistries and **lack the physical grounding** needed to handle unpredictable, real-world driving cycles [9].
3. We introduce a **physics-informed framework** that bridges this divide by using coupled electrical and thermal models to engineer unmeasurable core temperature ground-truth offline, uniquely empowering **lightweight algorithms** to achieve highly accurate, embedded-ready predictions.

To enable generalized training, we unified heterogeneous datasets (CALCE, NASA, Stanford, CALB) into a standardized, feature-extracted Parquet format.

Standardized Battery Degradation Datasets & Master Feature Set

Part A: Harmonized Experimental Datasets				
Data Source	**Cell Chemistries**	**Type**	**Raw Features (Time-Series)**	**Reference**
CALCE [1]	NMC, LFP	Cycle-Aging	Voltage, Current, Cell Temperature, Time	[1]
NASA [2]	LiCoO2, NCA, Other	Cycle-Aging	Voltage, Current, Surface Temperature, Time, capacity	[2]
Stanford [3]	LFP	Cycle-Aging, Cycle-Life	Voltage, Current, Environmental Temperature, Time	[3]
CALB [4]	NMC, LFP	Cycle-Aging	Voltage, Current, Surface Temperature, Time	[4]

Part B: Unified Master Parquet Dataset (This Study)			
Data Unification	**Extracted Cycle-Level Features**	**Engineered Physics-Informed Labels**	**Target Outputs**
Unified [1]-[4]	Statistical (Mean, Max, Min); Temporal (dV/dt); Spectral (FFT); Wavelet Transforms; Capacity Trends	Engineered Core Temperature (T_{core}); State of Health (SOH) Capacity Deviation	Predicted Core Temperature Predicted State of Health (SOH)

Fig 1. Battery Degradation Dataset and Master Parquet Dataset

Physics-Informed Electrical and Thermal Modeling

1. To bypass the physical impossibility of placing sensors inside a battery core, we engineered theoretical labels using a coupled electrical-thermal approach. A **second-order RC equivalent** circuit [5] was implemented to capture **polarization effects and internal resistance dynamics**.
2. The generated heat power $\mathbf{P_{loss}}$ was fed into a lumped-parameter thermal model [6]. Using the convective cooling rate ***b*** and heating gain ***a***, we accurately estimated the core temperature $\mathbf{T_{core}}$ from the measurable surface temperature $\mathbf{T_{ambient}}$.

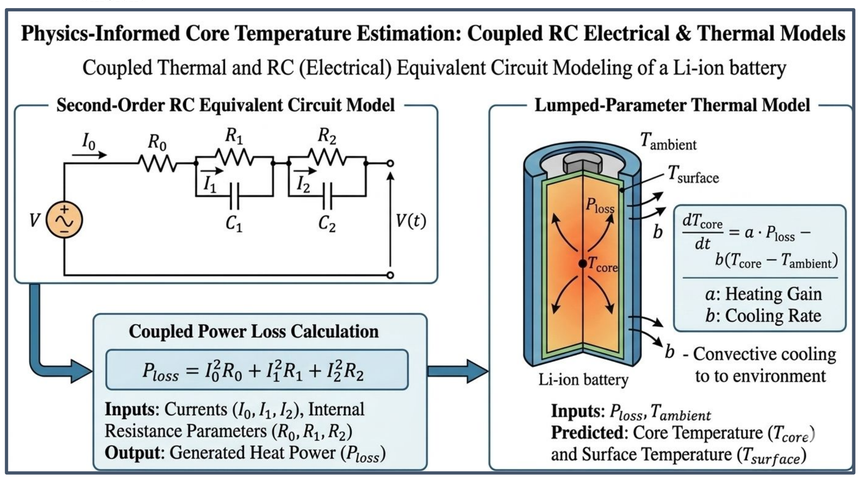

Fig 2. RC Electrical and Thermal Model of Battery

Supporting Equations[5][6]:

$$P_{loss} = I_0^2R_0 + I_1^2R_1 + I_2^2R_2$$

$$dT_{core}/dt = a.P_{loss} - b.(T_{core} - T_{ambient})$$

Feature Engineering and Predictive Architectures

1. We transformed raw time-series data into highly descriptive, cycle-level features. This included statistical **baselines, temporal dynamics (*dV/dt*), spectral transforms (FFT)**, and high-frequency wavelets. State of Health (SOH) was computed using capacity degradation trends across cycles.
2. To prevent data leakage, training pipelines utilized strict Group K Fold cross-validation grouped by cell ID. We evaluated a suite of machine learning models, including Decision Trees, K-Nearest Neighbors (KNN), Support Vector Regressors (SVR), Random Forests, MLP and XGBoost - specifically prioritized for low inference latency on embedded systems.

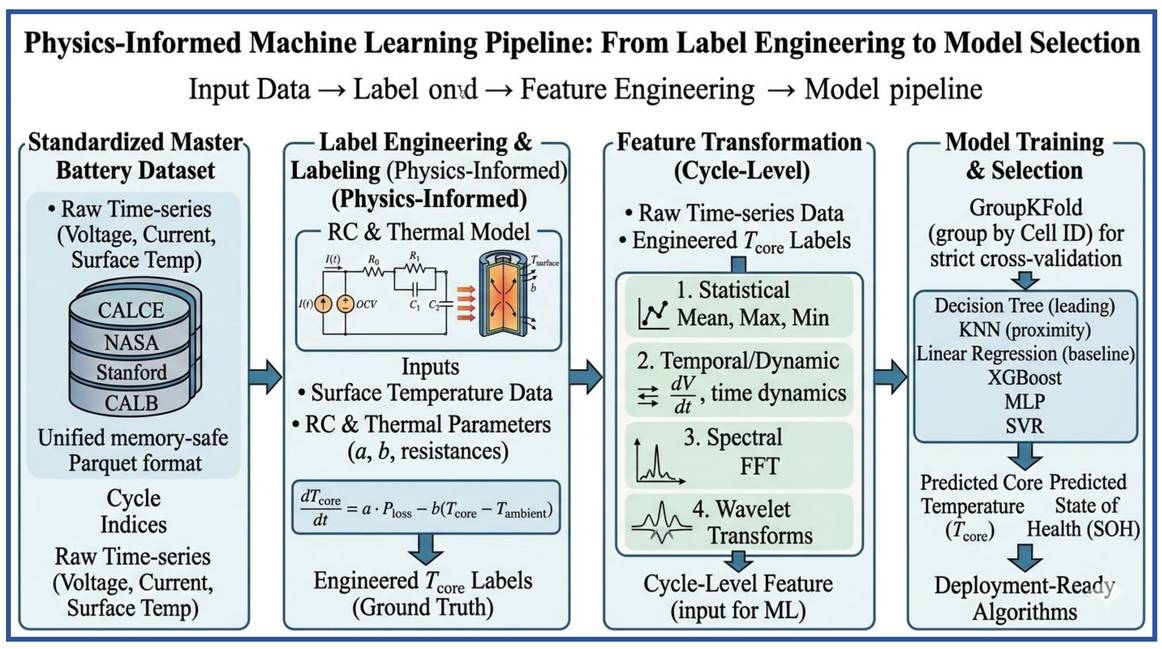

Fig 3. Complete Pipeline from labelling to ML Modelling

Performance Evaluation and Insights

Battery Core Temperature Prediction Results			
Machine Learning Model	**Root Mean Square Error (RMSE)**	**Mean Absolute Error (MAE)**	**R -Square Error (R2)**
RF	**0.14537**	**0.01901**	**0.9993**
XGBoost	**0.30409**	**0.1357**	**0.9969**
MLP	**0.17566**	**0.05684**	**0.999**
Linear Reg	**0.527739**	**0.23139**	**0.990628**
KNN	**0.164418**	**0.026687**	**0.99909**
Decision Tree	**0.06326**	**0.007124**	**0.999865**
SVR	**0.244495**	**0.080308**	**0.997988**
State of Health (SOH) Prediction Results			
Machine Learning Model	**Root Mean Square Error (RMSE)**	**Mean Absolute Error (MAE)**	**R -Square Error (R2)**
RF	**0.01379**	**0.00936**	**0.9977**
XGBoost	**0.02138**	**0.01705**	**0.9944**
MLP	**0.03664**	**0.02071**	**0.9836**
Linear Reg	**0.057431**	**0.039707**	**0.959754**
KNN	**0.028756**	**0.026687**	**0.98991**
Decision Tree	**0.024277**	**0.01394**	**0.959754**
SVR	**0.055871**	**0.046165**	**0.961911**

TABLE 1. ML Model Training Insights - RMSE, MAE and R^2 Errors

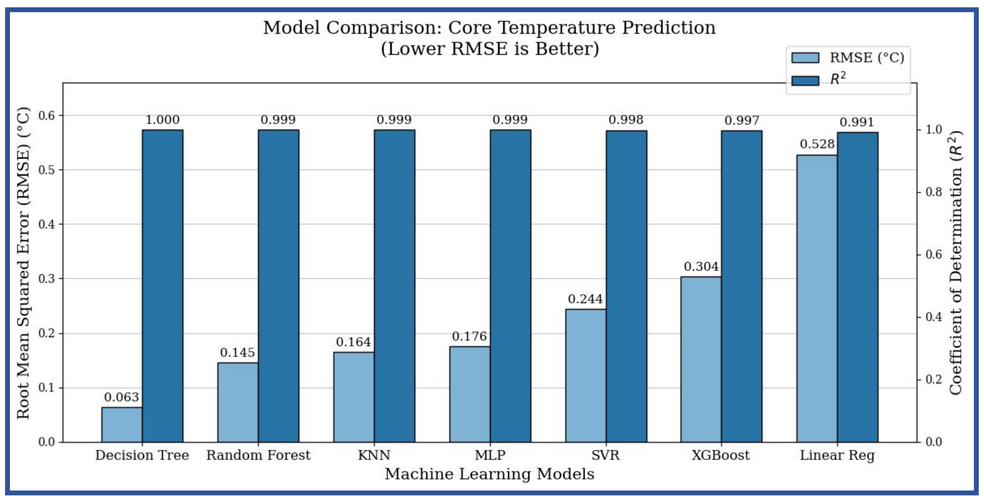

Fig 4. Model Comparison on Core Temperature Predictions

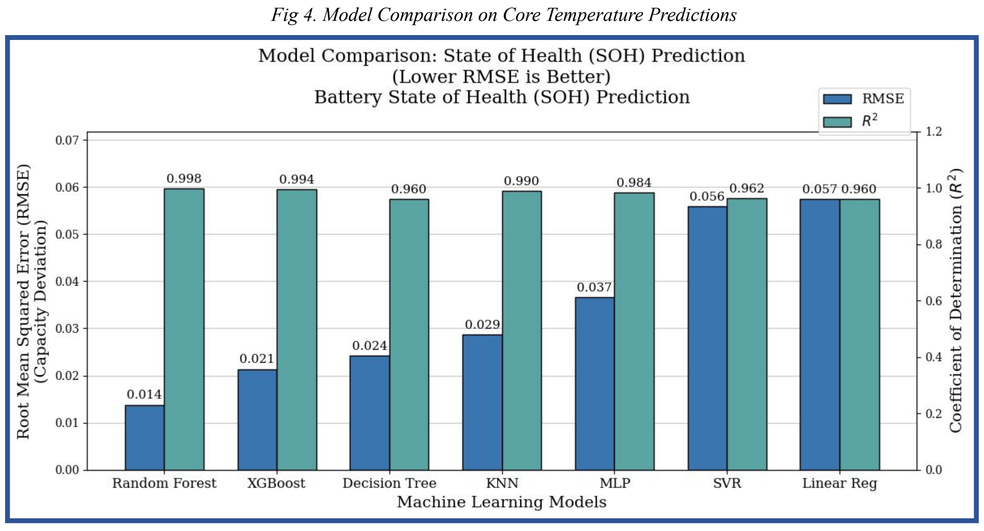

Fig 5. Model Comparison on State-of-Health Predictions

1. Tree-based and proximity models demonstrated exceptional capability in capturing non-linear thermal dynamics. The Decision Tree regressor outperformed all candidates, achieving an **RMSE of 0.063 and R^2 of 0.999** for core temperature prediction, while also leading SOH predictions.
2. Interestingly, heavier gradient boosters like **XGBoost and simple Linear Regression underperformed**. Linear models failed to capture the complex thermal inertia, while XGBoost over-penalized localized anomalies in the drive cycles, proving that highly localized, **non-linear interpolations (like Decision Trees) are optimally suited** for this specific feature space.

Future Scope and Timeline

1. **Phase 1: Data Harmonization & Physics-Informed Labeling (Completed)** Unified heterogeneous, multi-chemistry datasets (CALCE, NASA, Stanford, CALB) and engineered precise ground-truth core temperature $\mathbf{T_{core}}$ and State of Health (SOH) labels using coupled RC and thermal models.
2. **Phase 2: Algorithm Benchmarking & Optimization (Completed)** Trained and evaluated a diverse suite of machine learning models against our unified dataset, establishing the Decision Tree as the optimal champion algorithm (achieving a highly accurate 0.063 RMSE) for lightweight predictive modeling.
3. **Phase 3: Condition-Aware Embedded Deployment (Future Scope)** For transitioning this framework directly onto BMS microcontrollers, we will categorize real-time data (battery chemistry, ambient temperature bands, and terrain patterns) to deploy an explainable, hardware-optimized selection rule that dynamically balances RMSE accuracy against inference latency and memory constraints.

Acknowledgement

We extend our sincere gratitude to Prof. Pallavi Bharadwaj for providing this research opportunity under the **Advanced Transportation and Electrification Technology** course. Special thanks to the SPEL Lab for their continued support, guidance, and resources.

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